SmartShop Mobile Application

Mini Project Report

Project Title: SmartShop Mobile Application

Course Name: Mobile Application Development

Course Code: ITEC-343

Group ID: Group 1

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Introduction

In recent years, mobile applications have become an essential part of daily life. People use mobile apps for communication, education, entertainment, and shopping. Among these, shopping applications are very popular because they provide convenience and allow users to browse and purchase products easily without visiting physical stores.

This project presents the development of a mobile application called SmartShop using Flutter. The application simulates a simple shopping platform where users can view products, check their details, watch related videos, and access additional information. The project focuses on applying the concepts learned in the Mobile Application Development course in a practical way.

Problem Statement

Traditional shopping methods require time and effort, as users need to visit stores physically. In addition, customers may not have enough information about products before making a purchase decision.

The SmartShop application aims to solve these issues by providing a digital platform where users can:

- Browse products easily
- View detailed information
- Watch videos related to products
- Navigate between sections smoothly

Project Objectives

The main objectives of this project are as follows:

- To develop a mobile application using Flutter
- To design multiple interactive screens
- To display products in an organized and clear format
- To provide detailed information about each product
- To integrate video playback functionality

- To include a web or external content feature
- To create a contact page that includes student and course information
- To apply theoretical knowledge in a practical project

System Overview

The SmartShop application is divided into several main parts. Each part is implemented as a separate screen:

- Home Screen
- Products Screen
- Product Details Screen
- Video Screen
- WebView Screen
- Contact Us Screen

All screens are connected through navigation, allowing the user to move easily between them.

Application Design and Structure

The application is designed using a simple and modular structure. Each screen is written in a separate Dart file to make the code more organized and easier to manage.

Flutter widgets are used to build the user interface. The design focuses on clarity and ease of use, so the user can interact with the application without confusion.

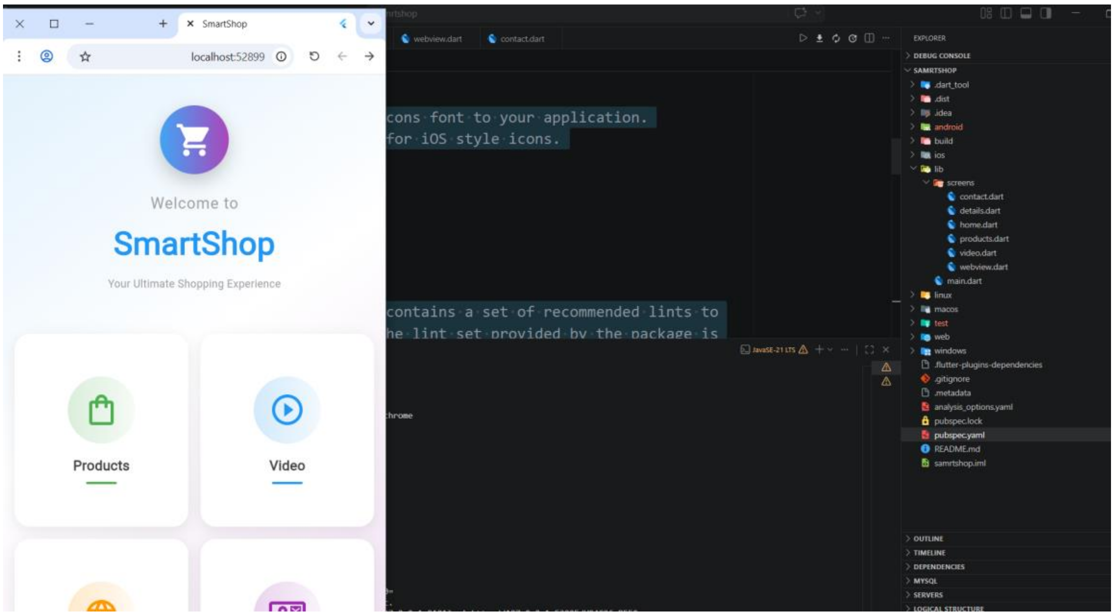
Navigation between screens is implemented using the Navigator class and MaterialPageRoute.

Screenshots of app interface

Home Screen

The Home Screen is the first screen that appears when the application starts. It contains a welcome message and buttons that allow the user to navigate to other screens such as products, videos, website, and contact page.

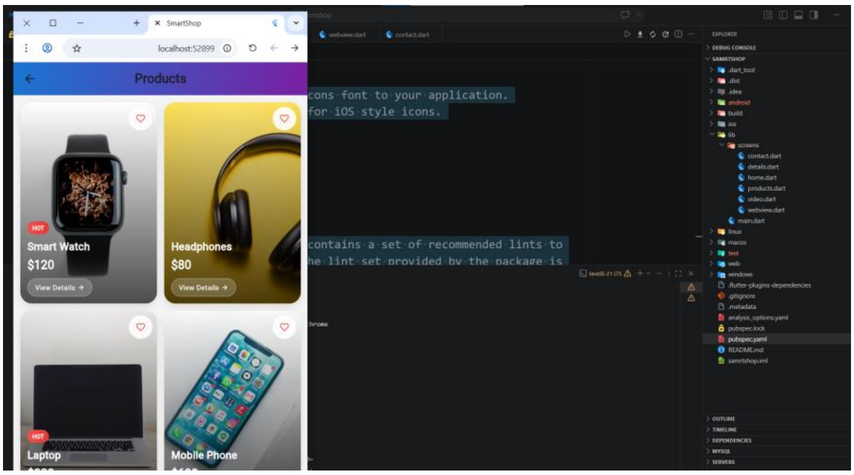
This screen serves as the main entry point of the application.



Products Screen

The Products Screen displays a list of products in a grid layout. Each product is shown with an image, name, and price.

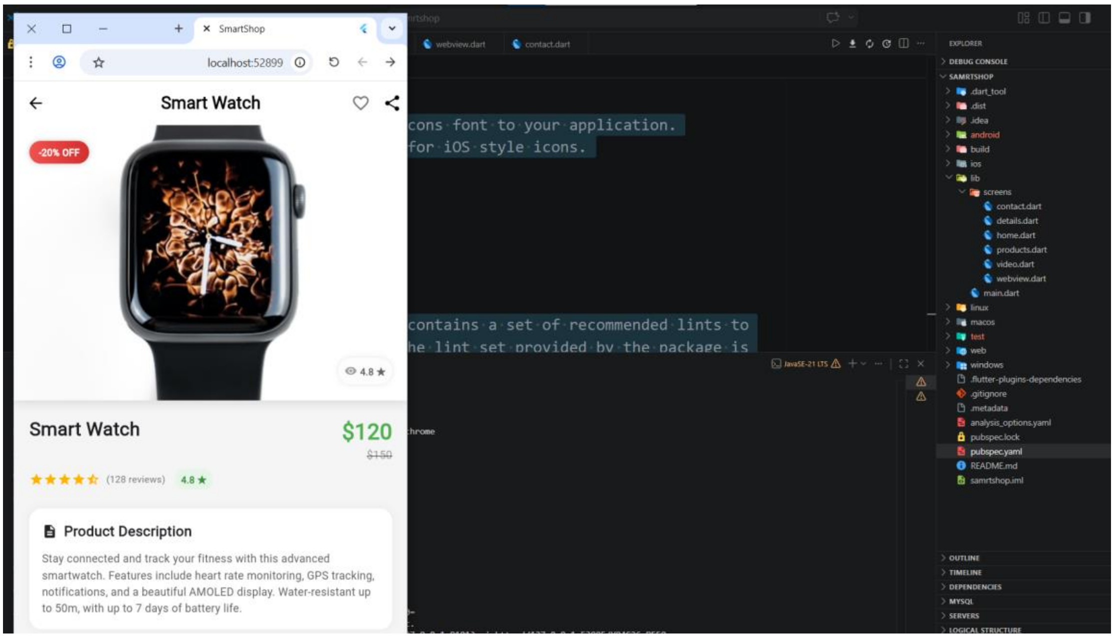
This screen allows the user to browse different products easily. When the user selects a product, they are taken to the details screen.



Product Details Screen

The Product Details Screen shows more information about the selected product. It includes a larger image, product name, price, and a short description.

There is also a button labeled "Buy Now" which simulates a purchase action.



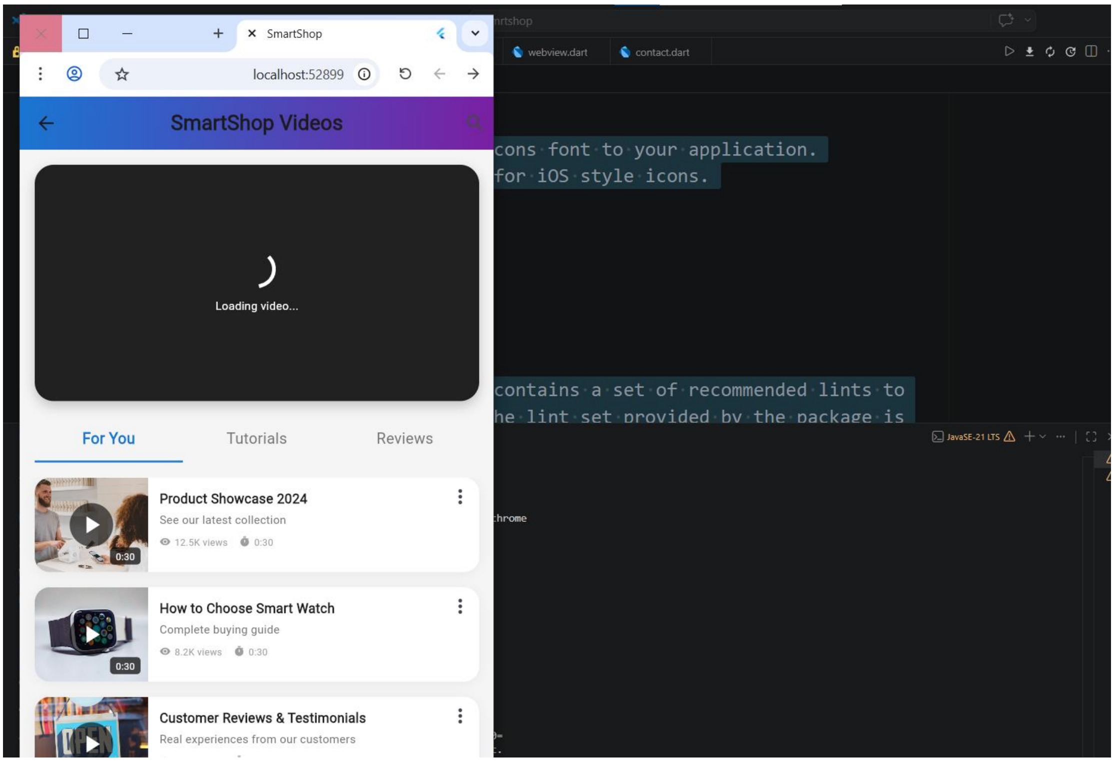
Video Screen

The Video Screen allows the user to watch videos related to the products. The videos are displayed using a video player.

The screen includes:

- A video player
- A loading indicator while the video is loading
- Play and pause controls
- A list of videos that the user can select

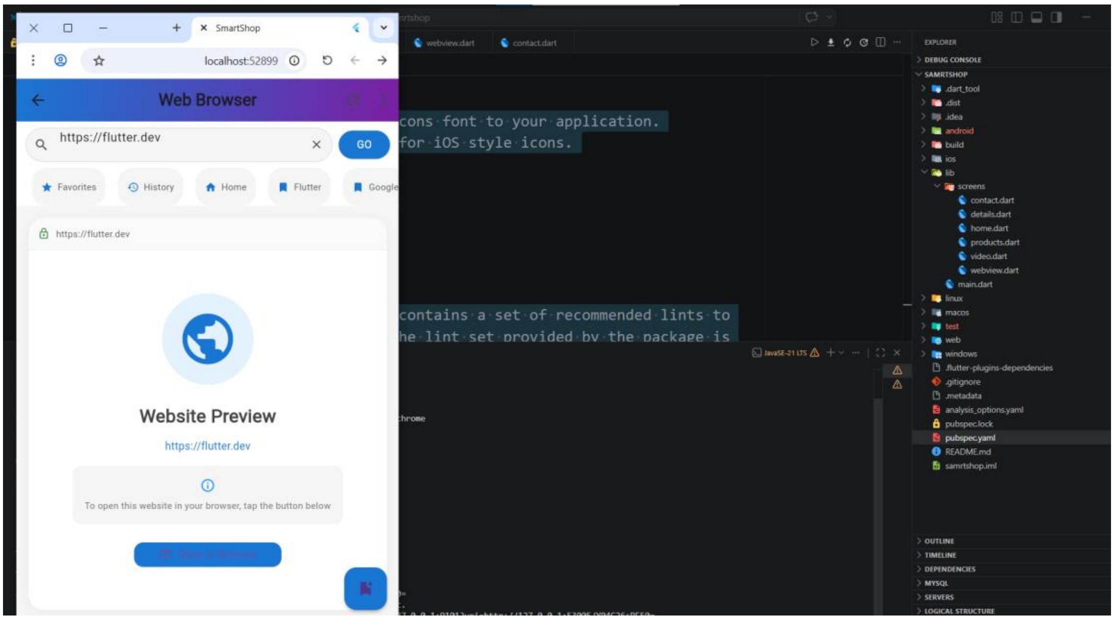
This feature improves user understanding of the products.



WebView Screen

The WebView Screen is used to display external web content. In some cases, due to environment limitations, a simplified version is used.

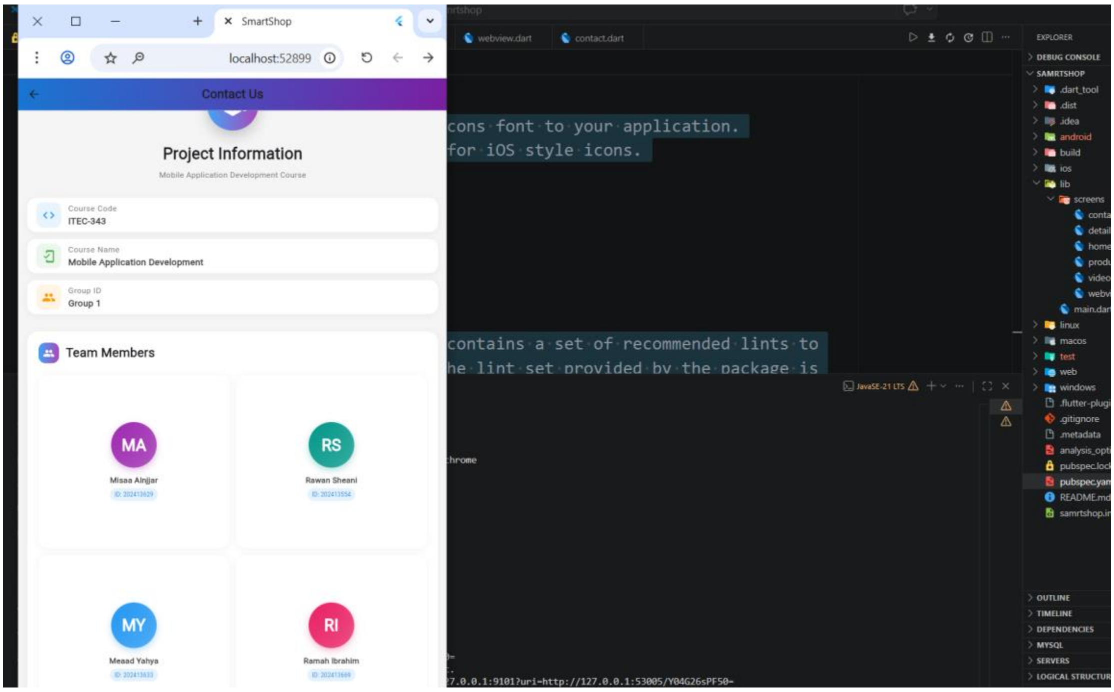
This screen demonstrates how mobile applications can integrate web-based information.



Contact Us Screen

The Contact Us Screen displays information about the students and the course. It includes names, student IDs, course name, and course code.

This screen is required as part of the project instructions.

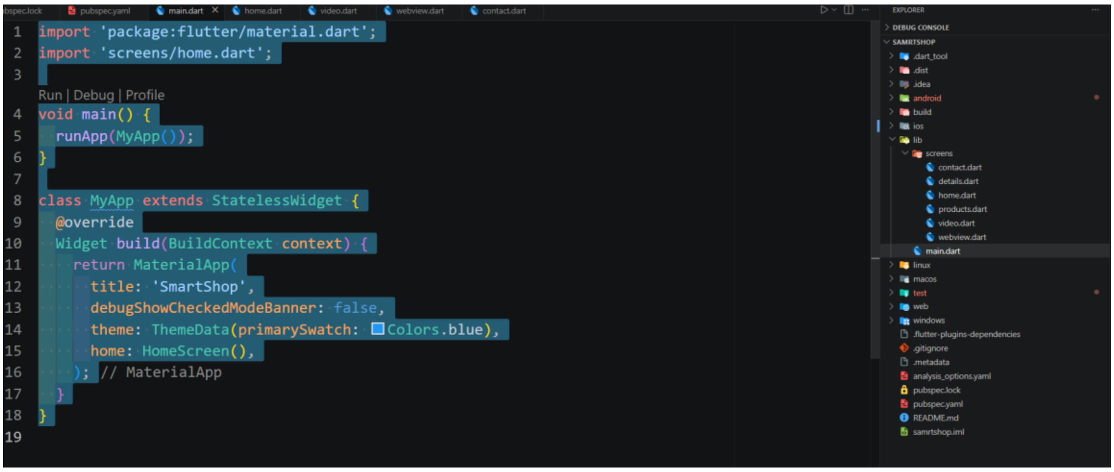


Technologies Used

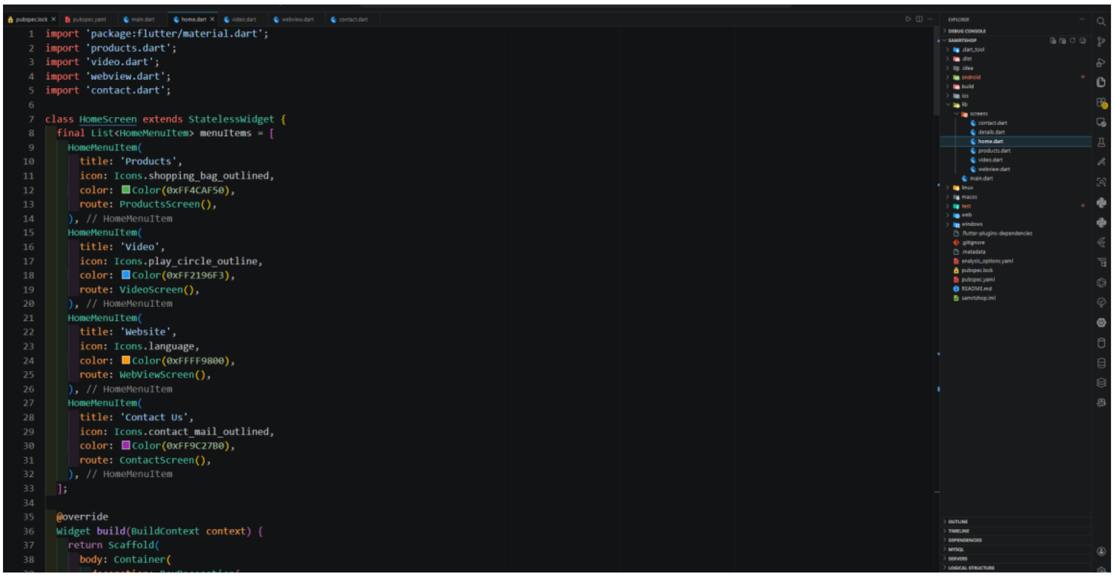
The application was developed using the following technologies:

- Flutter framework for building the user interface
- Dart programming language
- Material Design components
- video_player package for video playback

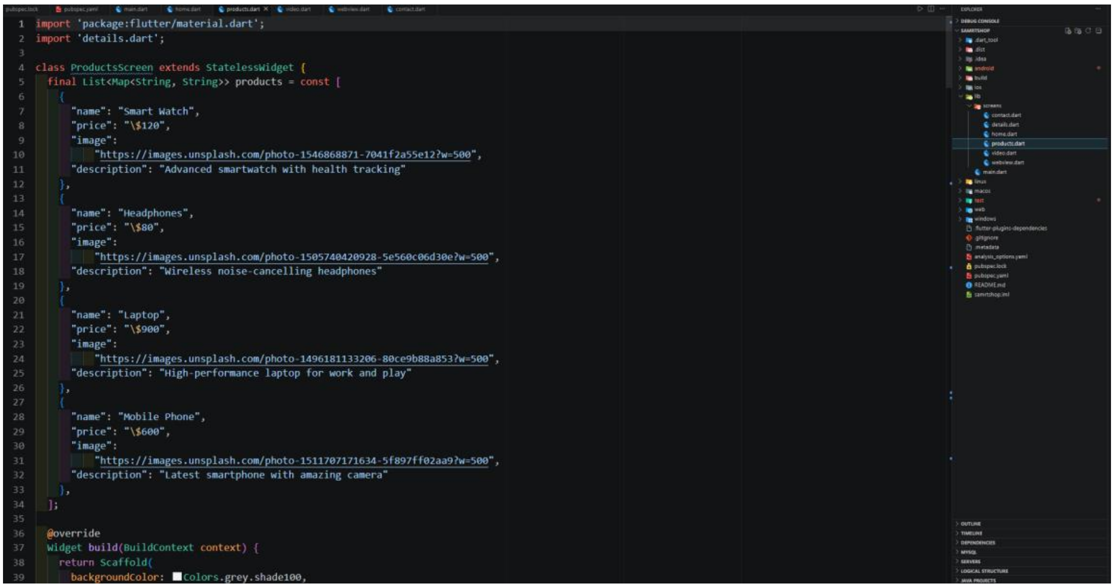
Code Snippets



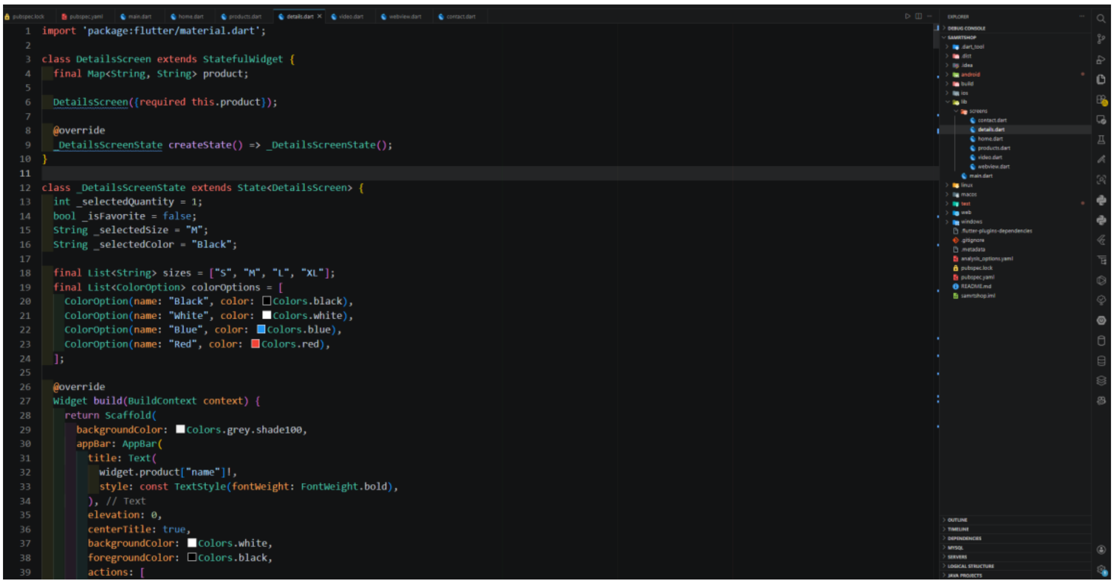
Home Page



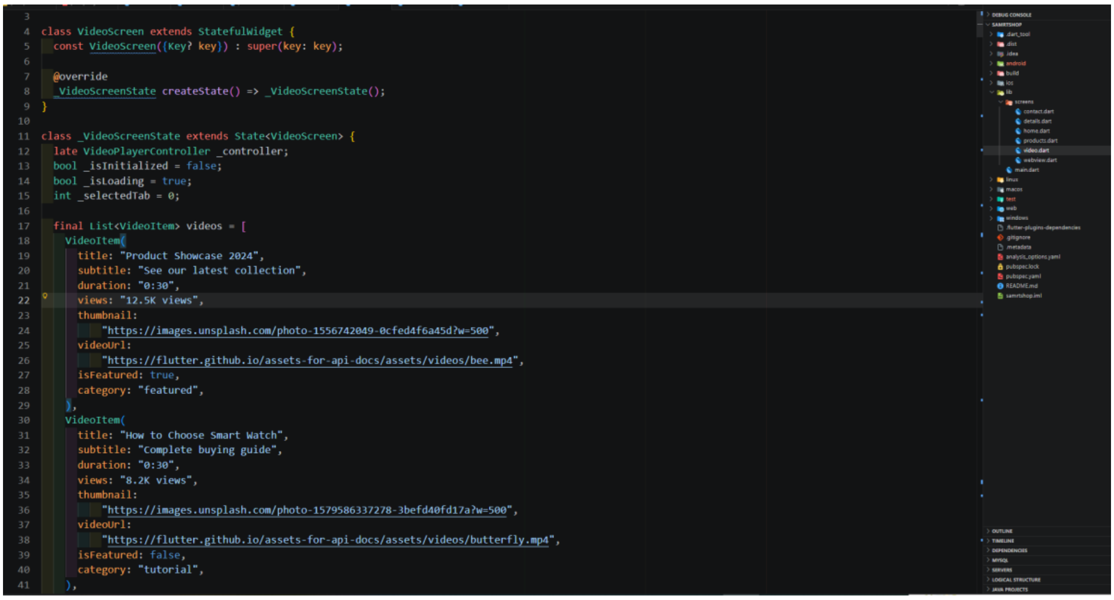
Products Page



Products Details Page

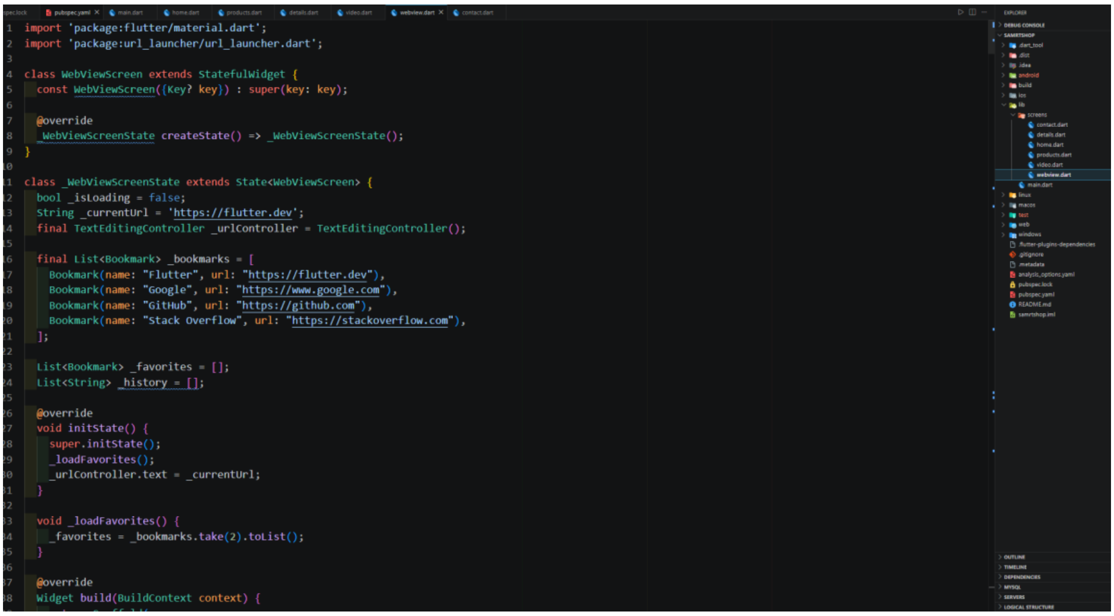


Video Page



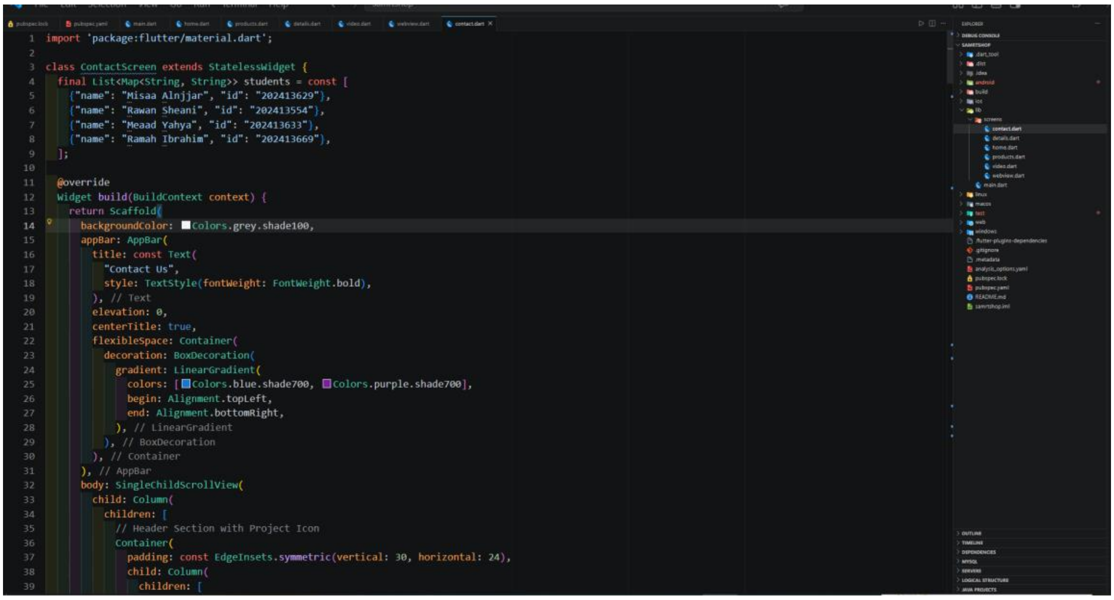
```dart
class VideoScreen extends StatefulWidget {
  const VideoScreen({Key? key}) : super(key: key);

  @override
  _VideoScreenState createState() => _VideoScreenState();
}

class _VideoScreenState extends State<VideoScreen> {
  late VideoPlayerController _controller;
  bool _isInitialized = false;
  bool _isLoading = true;
  int _selectedTab = 0;

  final List<VideoItem> videos = [
    VideoItem(
      title: "Product Showcase 2024",
      subtitle: "See our latest collection",
      duration: "0:30",
      views: "12.5K views",
      thumbnail:
          "https://images.unsplash.com/photo-1556742049-0cfed4f6a45d?w=500",
      videoUrl:
          "https://flutter.github.io/assets-for-api-docs/assets/videos/bee.mp4",
      isFeatured: true,
      category: "featured",
    ),
    VideoItem(
      title: "How to Choose Smart Watch",
      subtitle: "Complete buying guide",
      duration: "0:30",
      views: "8.2K views",
      thumbnail:
          "https://images.unsplash.com/photo-1579586337278-3befd40fd17a?w=500",
      videoUrl:
          "https://flutter.github.io/assets-for-api-docs/assets/videos/butterfly.mp4",
      isFeatured: false,
      category: "tutorial",
    ),
```

Webview Page



```dart
import 'package:flutter/material.dart';
import 'package:url_launcher/url_launcher.dart';

class WebViewScreen extends StatefulWidget {
  const WebViewScreen({Key? key}) : super(key: key);

  @override
  _WebViewScreenState createState() => _WebViewScreenState();
}

class _WebViewScreenState extends State<WebViewScreen> {
  bool _isLoading = false;
  String _currentUrl = 'https://flutter.dev';
  final TextEditingController _urlController = TextEditingController();

  final List<Bookmark> _bookmarks = [
    Bookmark(name: "Flutter", url: "https://flutter.dev"),
    Bookmark(name: "Google", url: "https://www.google.com"),
    Bookmark(name: "GitHub", url: "https://github.com"),
    Bookmark(name: "Stack Overflow", url: "https://stackoverflow.com"),
  ];

  List<Bookmark> _favorites = [];
  List<String> _history = [];

  @override
  void initState() {
    super.initState();
    _loadFavorites();
    _urlController.text = _currentUrl;
  }

  void _loadFavorites() {
    _favorites = _bookmarks.take(2).toList();
  }

  @override
  Widget build(BuildContext context) {
```

Contact us Page



Conclusion

In conclusion, the SmartShop application is a successful implementation of a simple mobile application using Flutter. The project helped in understanding how to design user interfaces, manage navigation, and integrate multimedia features.

It also provided practical experience in building a real application and solving development challenges.